ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 8.01: Integration & Kinematics

Q1. Evaluate the definite integral: int_0^1 2x + 1^3 , dx .

[ ] A: frac{80}{2}
[ ] B: frac{84}{8}
[ ] C: frac{80}{4}
[x] D: frac{80}{8}

Solution: Substitute u = 2x + 1 implies du = 2,dx . Integral: frac{1}{2}left[frac{u^4}{4}right]_1^{3} = frac{3^4 - 1^{4}}{8} = frac{80}{8} .

Q2. Evaluate the definite integral: int_0^1 3x + 1^3 , dx .

[ ] A: frac{255}{4}
[ ] B: frac{255}{3}
[ ] C: frac{259}{12}
[x] D: frac{255}{12}

Solution: Substitute u = 3x + 1 implies du = 3,dx . Integral: frac{1}{3}left[frac{u^4}{4}right]_1^{4} = frac{4^4 - 1^{4}}{12} = frac{255}{12} .

Q3. Evaluate the definite integral: int_0^1 4x + 1^3 , dx .

[ ] A: frac{625}{16}
[ ] B: frac{624}{4}
[ ] C: frac{628}{16}
[x] D: frac{624}{16}

Solution: Substitute u = 4x + 1 implies du = 4,dx . Integral: frac{1}{4}left[frac{u^4}{4}right]_1^{5} = frac{5^4 - 1^{4}}{16} = frac{624}{16} .

Q4. Evaluate the definite integral: int_0^1 5x + 1^3 , dx .

[ ] A: frac{1295}{5}
[ ] B: frac{1295}{4}
[x] C: frac{1295}{20}
[ ] D: frac{1299}{20}

Solution: Substitute u = 5x + 1 implies du = 5,dx . Integral: frac{1}{5}left[frac{u^4}{4}right]_1^{6} = frac{6^4 - 1^{4}}{20} = frac{1295}{20} .

Q5. Evaluate the definite integral: int_0^1 6x + 1^3 , dx .

[ ] A: frac{2400}{6}
[x] B: frac{2400}{24}
[ ] C: frac{2404}{24}
[ ] D: frac{2400}{4}

Solution: Substitute u = 6x + 1 implies du = 6,dx . Integral: frac{1}{6}left[frac{u^4}{4}right]_1^{7} = frac{7^4 - 1^{4}}{24} = frac{2400}{24} .

Q6. Evaluate the definite integral: int_0^1 7x + 1^3 , dx .

[ ] A: frac{4095}{4}
[ ] B: frac{4099}{28}
[x] C: frac{4095}{28}
[ ] D: frac{4095}{7}

Solution: Substitute u = 7x + 1 implies du = 7,dx . Integral: frac{1}{7}left[frac{u^4}{4}right]_1^{8} = frac{8^4 - 1^{4}}{28} = frac{4095}{28} .

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Q7. Evaluate the definite integral: int_0^1 8x + 1^3 , dx .

[ ] A: frac{6560}{4}
[ ] B: frac{6560}{8}
[ ] C: frac{6564}{32}
[x] D: frac{6560}{32}

Solution: Substitute u = 8x + 1 implies du = 8,dx . Integral: frac{1}{8}left[frac{u^4}{4}right]_1^{9} = frac{9^4 - 1^{4}}{32} = frac{6560}{32} .

Q8. Evaluate the definite integral: int_0^1 9x + 1^3 , dx .

[ ] A: frac{9999}{4}
[ ] B: frac{9999}{9}
[ ] C: frac{10003}{36}
[x] D: frac{9999}{36}

Solution: Substitute u = 9x + 1 implies du = 9,dx . Integral: frac{1}{9}left[frac{u^4}{4}right]_1^{10} = frac{10^4 - 1^{4}}{36} = frac{9999}{36} .